

Leader in the innovation of drug development for age-related diseases



One Platform, Dual Solutions:

Innovative mTORC2 Inhibitor Design Technology for Sarcopenia and Alzheimer's Disease Treatments

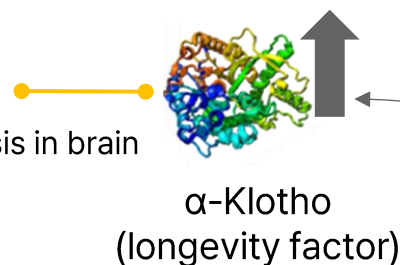
Solving two key aging challenges with a single platform technology

the main causes of aging

the mechanism targeting the main causes of aging

[Longevity factor] Decrease of α -Klotho

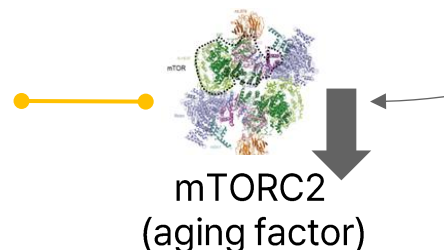
- A longevity hormone which continuously decreases with age
- α -Klotho is necessary for muscle regeneration and neurogenesis in brain
- It's decline contributes to the cause of sarcopenia and AD
- α -Klotho improves the age-related diseases in animal models



mTORC2 and α -Klotho
mutual negative regulators

[Aging factor] mTORC2 hyperactivation

- A key factor of aging and lifespan from C. elegans to humans
- A major cause of sarcopenia and Alzheimer's disease



Increase

Inhibition

mTORC2 inhibitor



Dual-Targeting Platform Technology

- Elimination of the causes of sarcopenia and AD
- Rewinding the aging clock through α -Klotho

We are developing ...

Therapeutic agent for sarcopenia

- Core technology
 - Selective mTORC2 PPI inhibitor
- Key differentiations
 - A selective mTORC2 PPI inhibitor that suppresses the overactivation of mTORC2, the underlying cause of sarcopenia
 - A drug that promotes muscle regeneration by increasing α -Klotho
 - A drug that increases the size of skeletal muscle fibers that decrease with aging
 - A drug that improves muscle strength and physical performance abilities that decline with aging
 - Oral medication

Therapeutic agent for Alzheimer's disease

- Core technology
 - Selective mTORC2 PPI inhibitor
- Key differentiations
 - A selective mTORC2 PPI inhibitor that suppresses the overproduction of $A\beta$
 - A drug that restore damaged neural networks by increasing α -Klotho
 - A drug that reduces amyloid plaques and phospho-tau in brain while suppressing brain inflammation
 - A drug that improves cognitive function in Alzheimer's disease patients
 - Oral medication

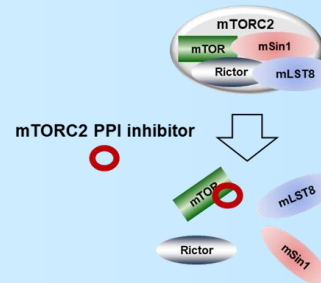
Development of Selective mTORC2 Inhibitors

No selective mTORC2 inhibitors available

- mTORC2 shares mTOR, the catalytic subunit, with mTORC1
- Allosteric mTOR inhibitors and ATP-competitive mTOR inhibitors inhibit both mTORC1 and mTORC2



Inhibitors targeting only mTORC2
without affecting mTORC1

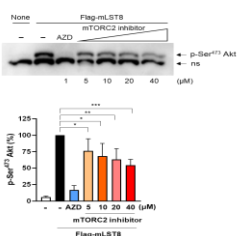


mTORC2 PPI inhibitor

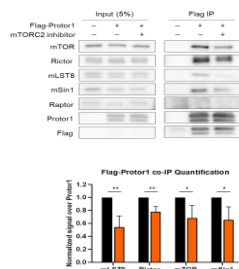
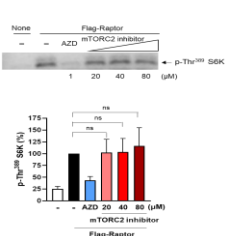
- Small molecule: MW < 500 Da
- Excellent BBB permeability

Selective inhibition of mTORC2

in vitro mTORC2 kinase assay

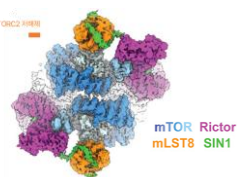


in vitro mTORC1 kinase assay



Break the interaction interface of the target protein with mTOR

mTORC2



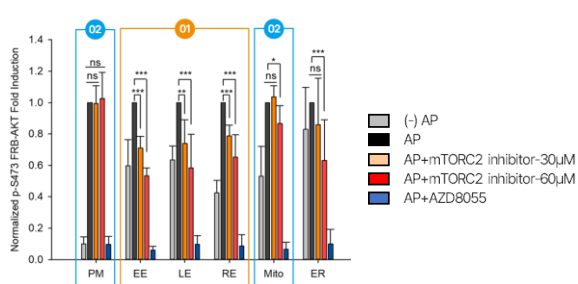
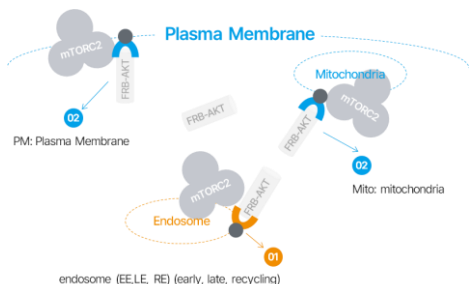
mTOR-the target protein MD simulation



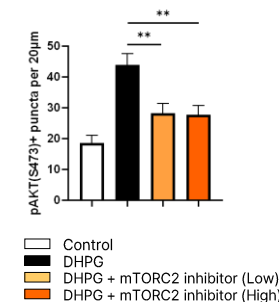
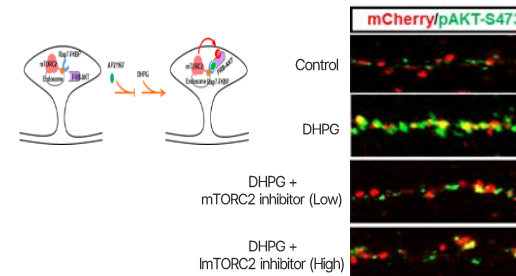
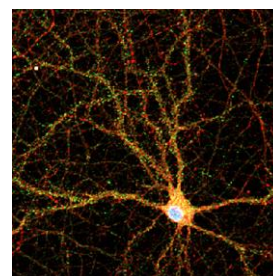
CDOCKER INTERACTION ENERGY
-44.6028 kcal/mol

MD simulation	Average interface distance, Å
mTORC2-Apo	4.61±0.62
mTORC2-inhibitor	9.68±2.80

Subcellular compartment(endosomes)-selective inhibition



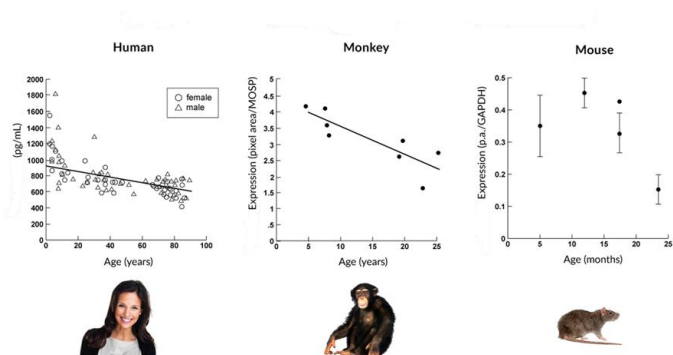
Inhibition of mTORC2 at endosomes of dendritic spines





Rewinding the aging clock: mTORC2 inhibitor promotes α -Klotho expression, a longevity factor

The decline of α -Klotho with age

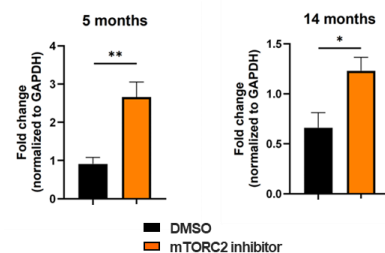


- $KL^{-/-}$ mice : premature aging phenotypes including muscle atrophy
- α -Klotho: Enhancement of memory in aged nonhuman primates

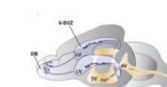


Promotion of α -Klotho expression by mTORC2 inhibitor

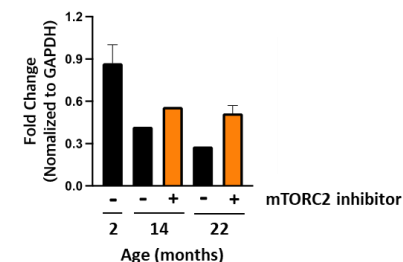
- Increase of α -Klotho mRNA (normal mice, lower limb muscle)



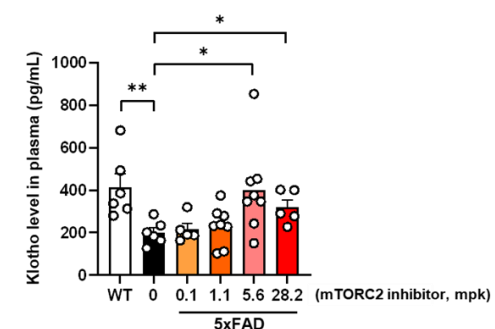
- Increase of α -Klotho mRNA (normal mice, choroid plexus)



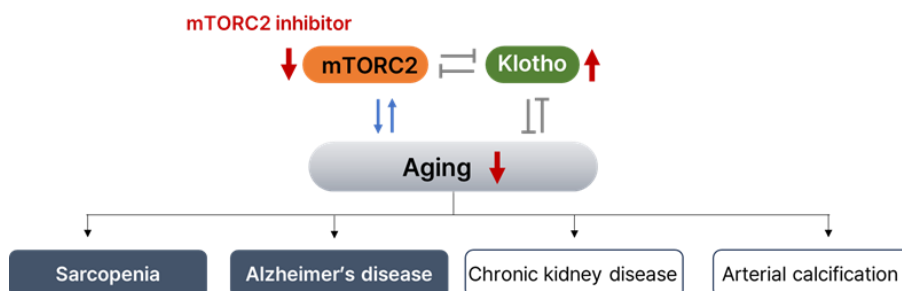
Klotho mRNA in Choroid plexus



- Increase of α -Klotho protein (6M 5xFAD mice, blood)



α -Klotho and mTOR signaling : mutual negative regulators





Sarcopenia

Pioneering a blue ocean in the sarcopenia market with our first-in-class treatment

Global market for sarcopenia treatments

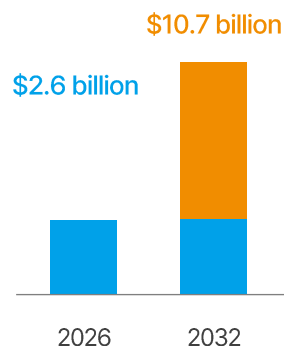
- **Prevalence: 10-26% of adults aged 60 and older**
- **Patient: Over 150 million worldwide, with a steady increase**
- **No FDA-Approved Treatments**

Growing Demand

- Entry into a super-aged society
- Growing awareness of sarcopenia

Global market for Sarcopenia treatments

- 2026 : **\$2.6 billion USD**
- CAGR : **26.5%**



Status of competitive drug development

Company	Component	MoA	Clinical trial	Status
Eli Lilly	Bimagrumab	ACVR2B antibody	Phase II	Ongoing
Regeneron	Trevogrumab + Garetosmab	Myostatin antibody + ACVR2A antibody	Phase II	Ongoing
TNF Pharm	MYMD1	TNF-α inhibitor	Phase II	Completed
Biophytis SA	BIO101 (20-hydroxy ecdysone)	MAS receptor agonist	Phase II	Completed
Aventi Biotechnology	AVTR101 (Alverine citrate)	Smooth muscle relaxation	Phase II	Completed
Animuscure	AMC6156	NSAIDs	Phase II	Ongoing
KSB Tugen	KSB-10301	?	Phase II	Ongoing

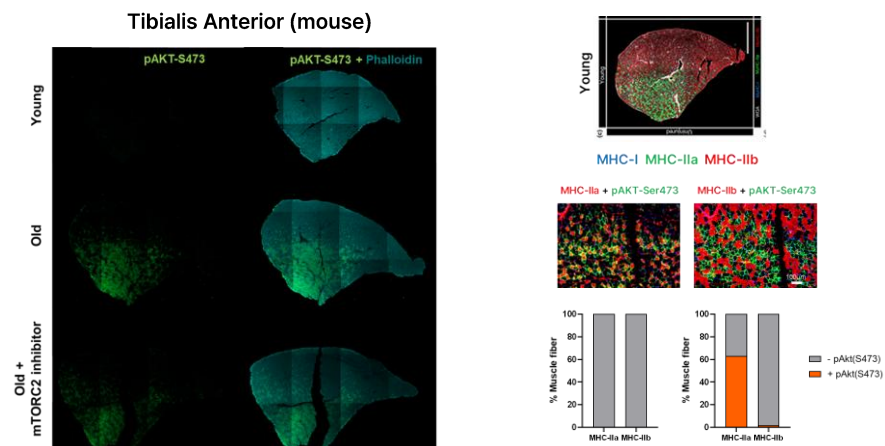
No causal treatment available

➤ **Urgent needs to develop causal treatments**

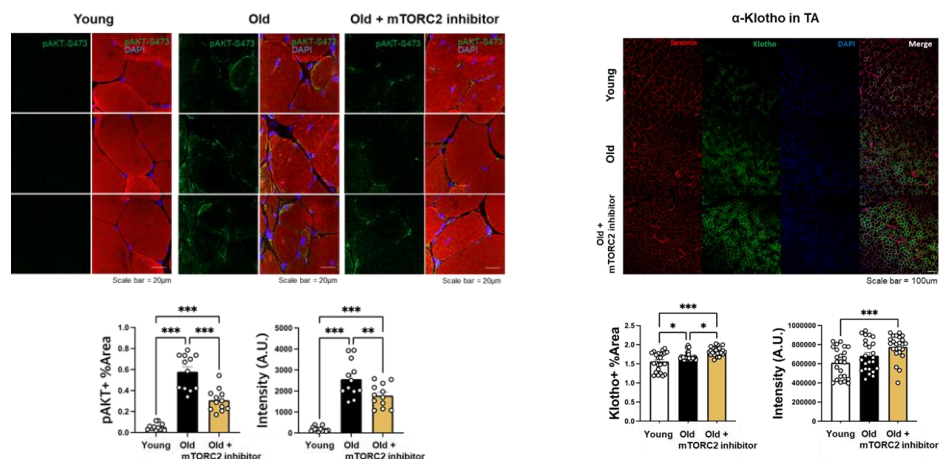


Causes of Sarcopenia: mTORC2 Overactivation

Overactivation of mTORC2 in aged skeletal muscle

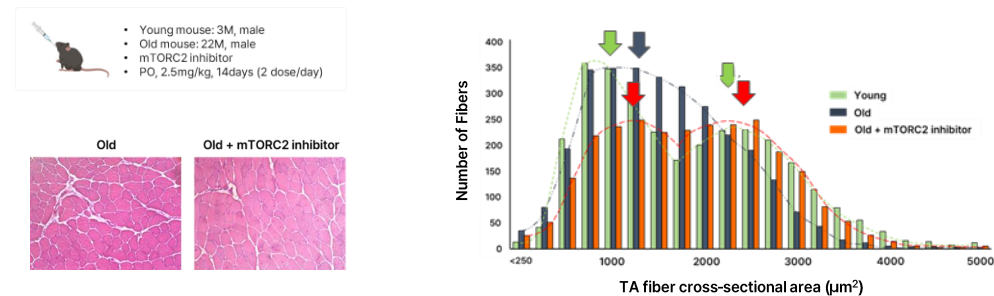


Suppression of mTORC2 and Increase of α -Klotho

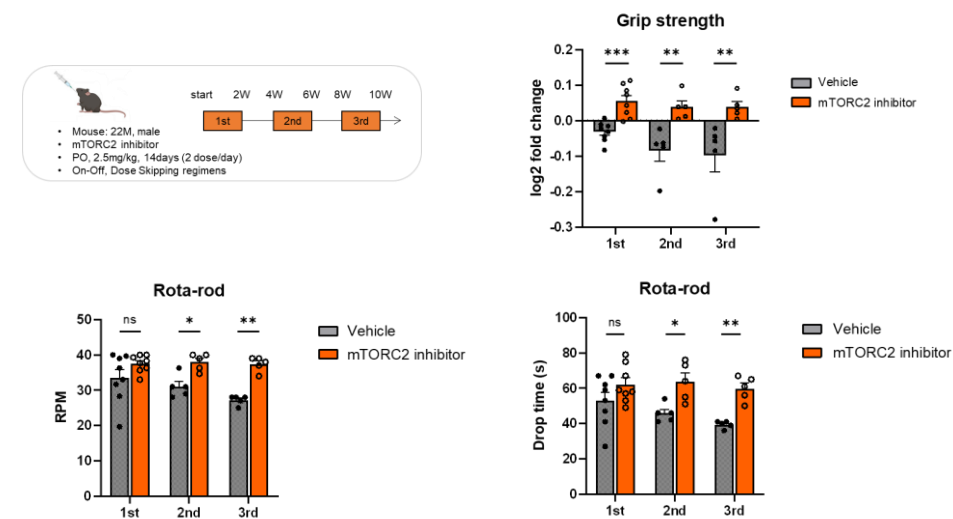


Improvement of Muscle Strength and Function in Aged Mice by an mTORC2 Inhibitor

Increase in the fiber cross-section area

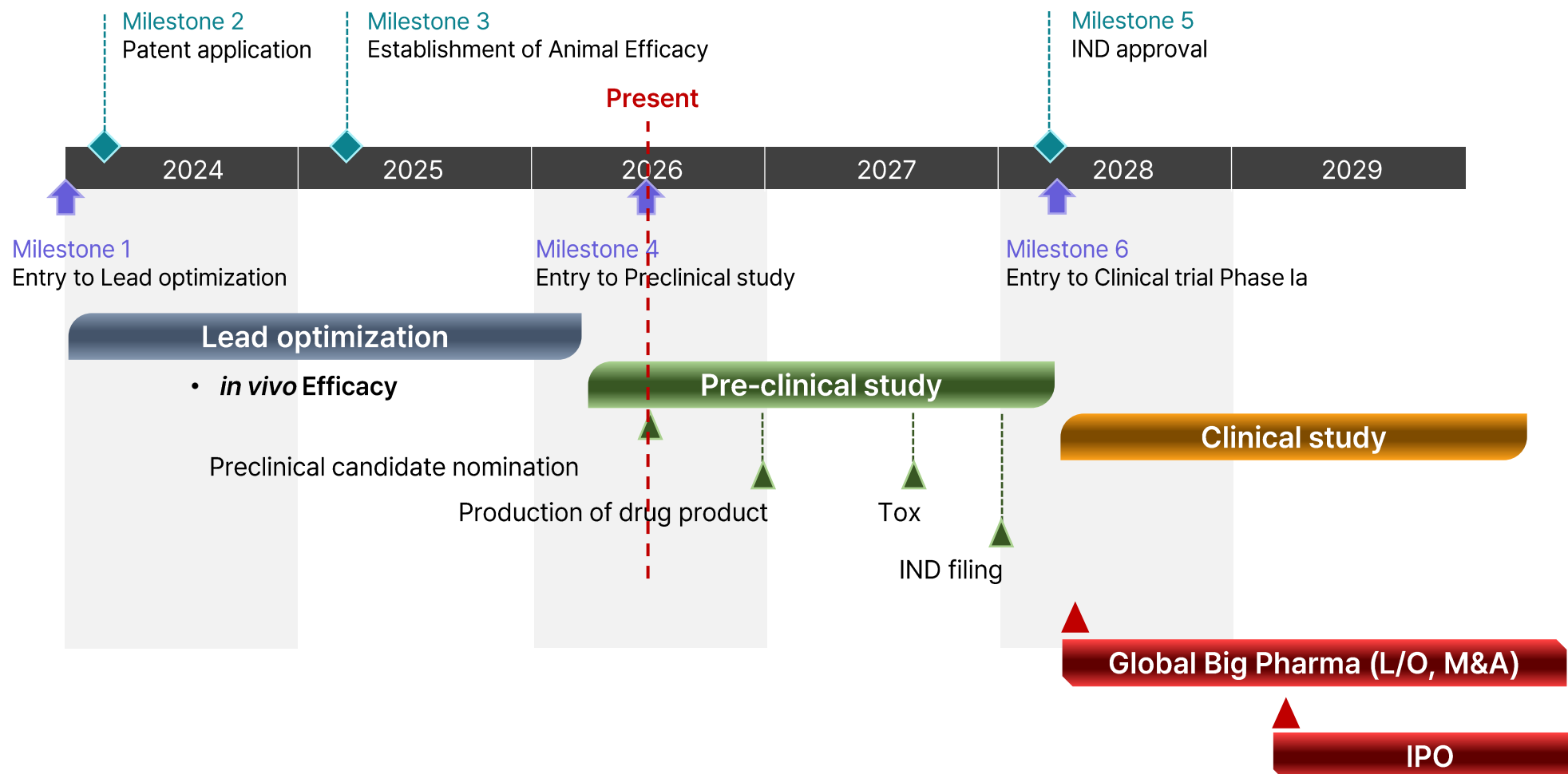


Improvement in muscle strength and functions





Entry of sarcopenia therapeutic candidate into the non-clinical (GLP-Tox) stage



Scale Up Strategy

Leveraging the fact that mTORC2 hyperactivation is a key underlying cause of aging-related diseases, we can build extensive pipelines for aging-related diseases

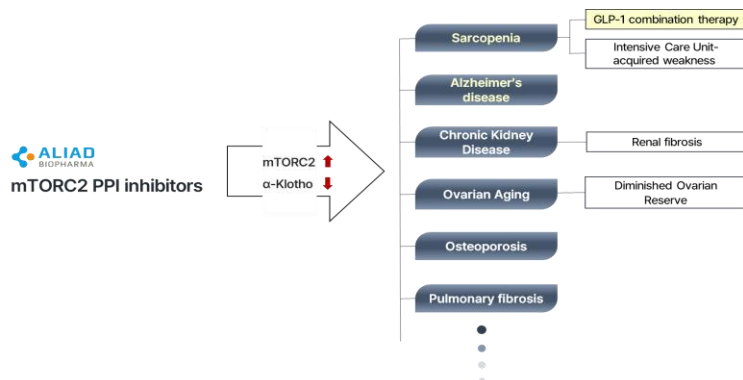
● Pipelines

Aliad Biopharma						
Lead Candidate	Indication	Discovery	Optimization	Preclinical	Clinical	Orphan Designation
AB-SP-001	Sarcopenia					
AB-AD-002	Alzheimer's Disease					
AB-AS-003	Autism Spectrum Disorder					● for PTEN hamartoma tumor syndrome (PHTS)
AB-SP-004	Muscle-preserving combination therapies w/ GLP-1 receptor agonists	PoC				

Indication Expansion

➤ Common Causative Factors for Multiple Age-related Diseases (Over-activation of mTORC2 & Decline in α-Klotho)

→ Unlocking it's Vast Indication Expansion Potential



Combination therapy with obesity drugs

➤ Combination therapy to address muscle loss, a side effect of GLP-1 obesity medications

- Myostatin and ACVR2 antibodies: Mitigate muscle loss induced by GLP-1 class obesity drugs
- α-Klotho: Blocks myostatin and activin receptor

➤ We are currently conducting a PoC to validate this approach



Alzheimer's disease

Urgent needs for innovative new drugs blocking A β overproduction

Global market for AD treatments

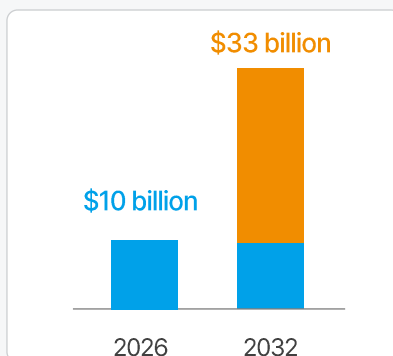
- Prevalence: 10% of adults aged 65 and older, 40% of adults aged 85 and older.
- Patient Population: Over 55 million dementia patients worldwide
- Alzheimer's Disease Proportion: 70% of all dementia patients

Growing Demand

- Entry into a super-aged society
- Paradigm shift after antibody drug approval
- Surge in demand for prevention driven by the ability to predict disease onset

Global market for AD treatments

- 2026: \$10 billion USD
- CAGR: 22.1%



Status of competitive drug development

Company	Component	MoA	Clinical trial	Status
Eisai/Biogen	Leqembi	Amyloid antibody		On market
Eli Lilly	Kisunla	Amyloid antibody		On market
AriBio	AR1001	PDE inhibitor	Phase III	Ongoing

Limitations of antibody drugs

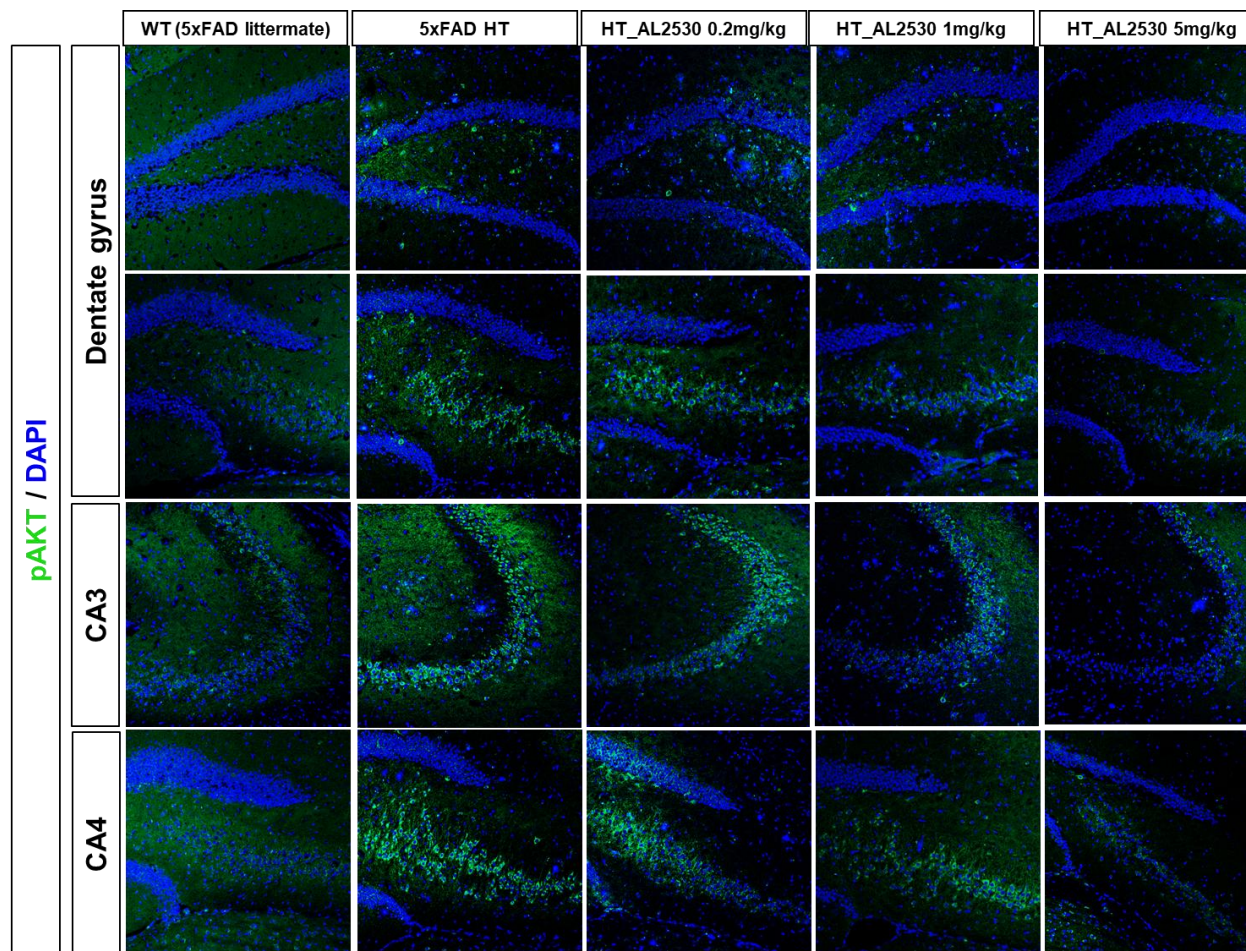
Category	Antibody drugs
Root cause of pathogenesis	Unable to block excessive A β production
Efficacy	Limited delaying effect
Recovery of neural networks	Unable
Side effects	High (Brain edema, Brain hemorrhage)
Route of administration	IV (one a month: Kisunla; twice a month: Leqembi)
Costs	Very high (US\$ 25,000~30,000 /year)

No drug with therapeutic efficacy

➤ Urgent needs to develop drugs blocking A β overproduction



- Over-activation of mTORC2 in the brain of 5xFAD mouse
- Suppression of mTORC2 by AL-2530 in the brain of 5xFAD mouse

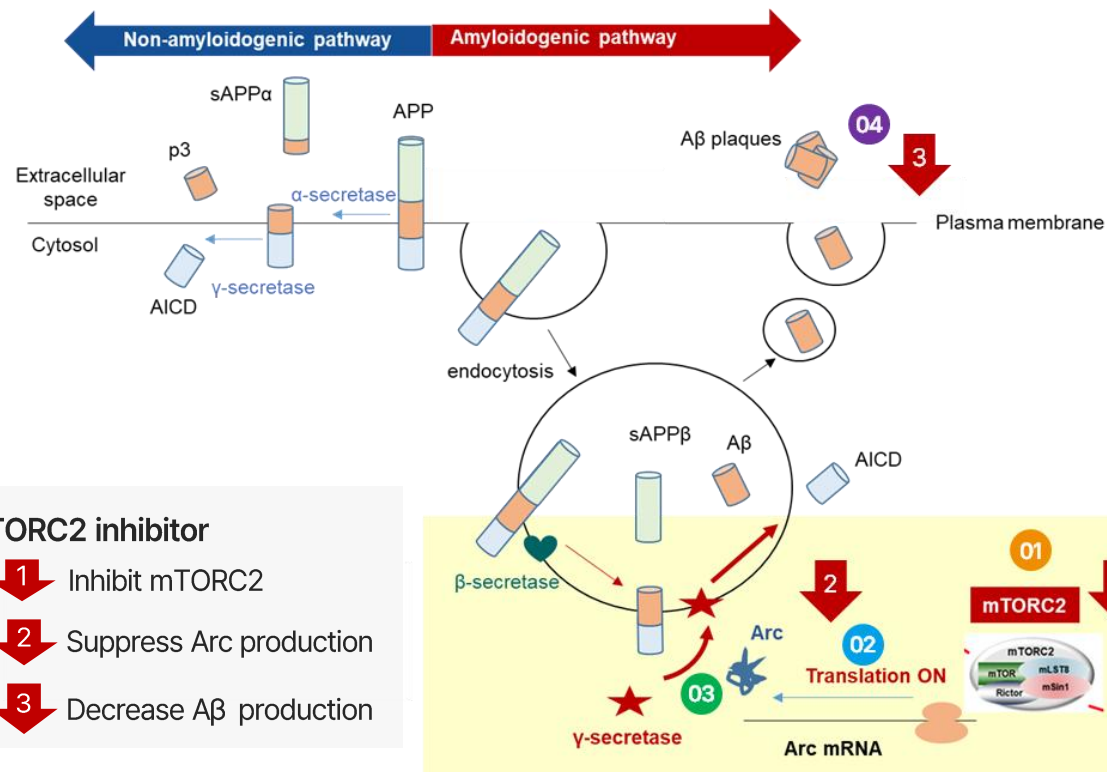


- 5xFAD mouse, 6M, Female
- mTORC2 inhibitor: AL-2530
- PO, 14 days



mTORC2 inhibitors block excessive A β production

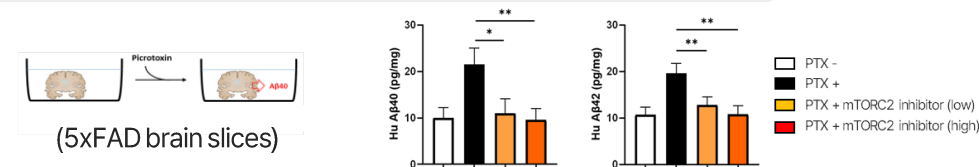
- 01 In brain, mTORC2 is activated gradually with aging
- 02 The activation of mTORC2 promotes the production of Arc protein
- 03 Arc recruits γ -secretase to endosomes, leading to excess production of A β
- 04 The excess A β forms aggregates in the brain, leading to Alzheimer's disease



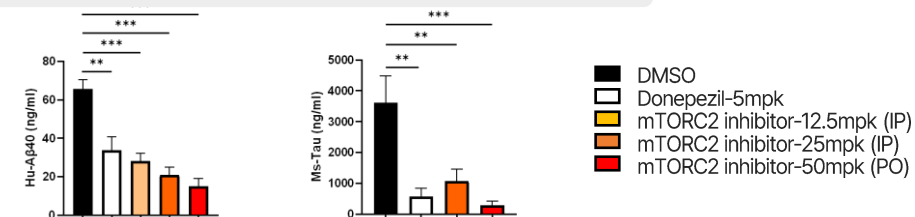
mTORC2 inhibitor

- 1 Inhibit mTORC2
- 2 Suppress Arc production
- 3 Decrease A β production

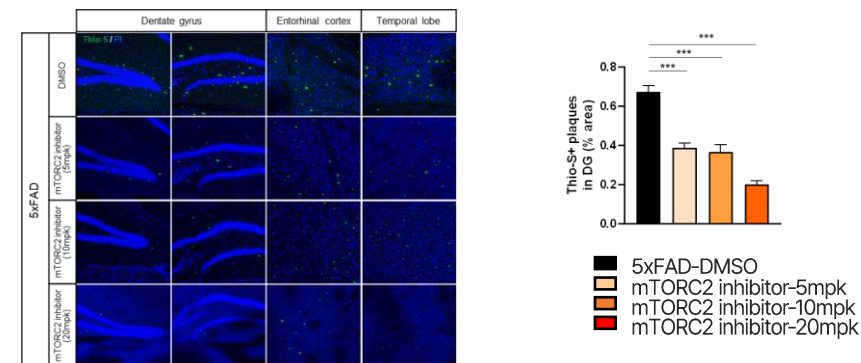
Suppression of the activity-dependent A β production



Decrease of A β and p-Tau in CSF of 5xHAD mice



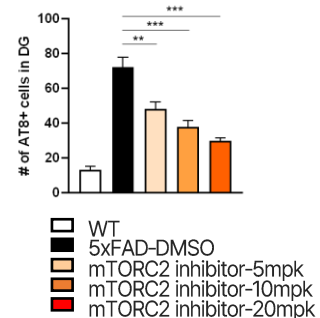
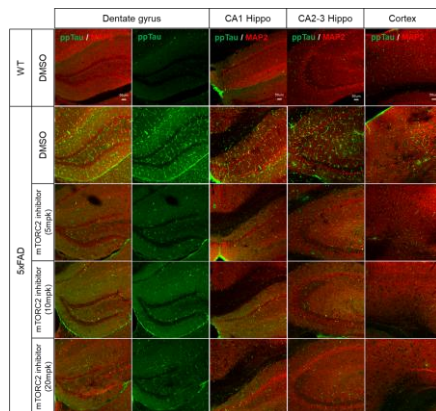
Decrease of amyloid plaque of 5xHAD mice



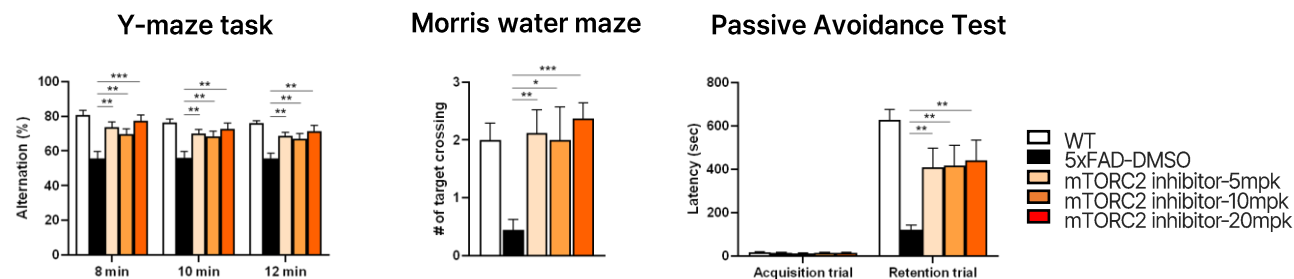


mTORC2 inhibitors restore damaged neural networks and improve cognitive function

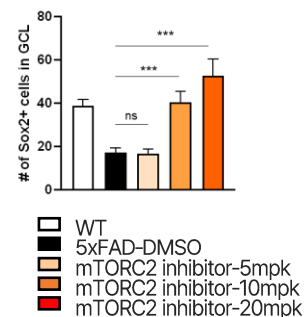
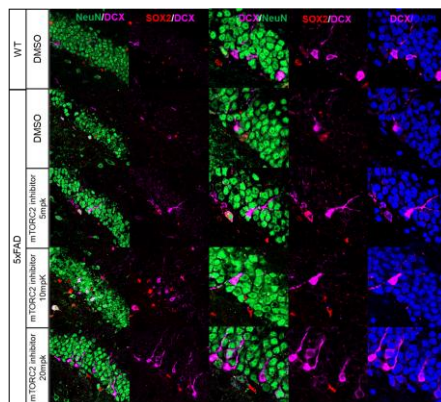
Decrease of phospho-Tau



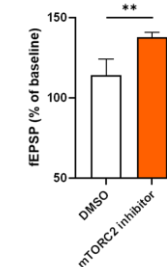
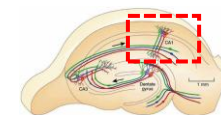
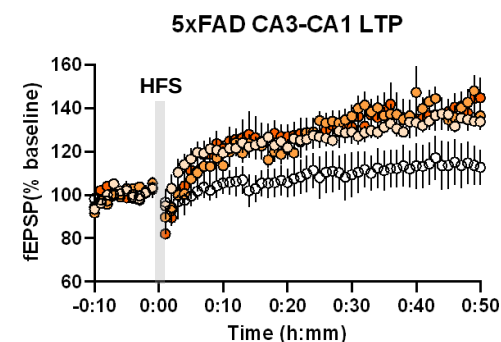
Rescue of AD-related cognitive impairments



Promotion of neurogenesis in hippocampus

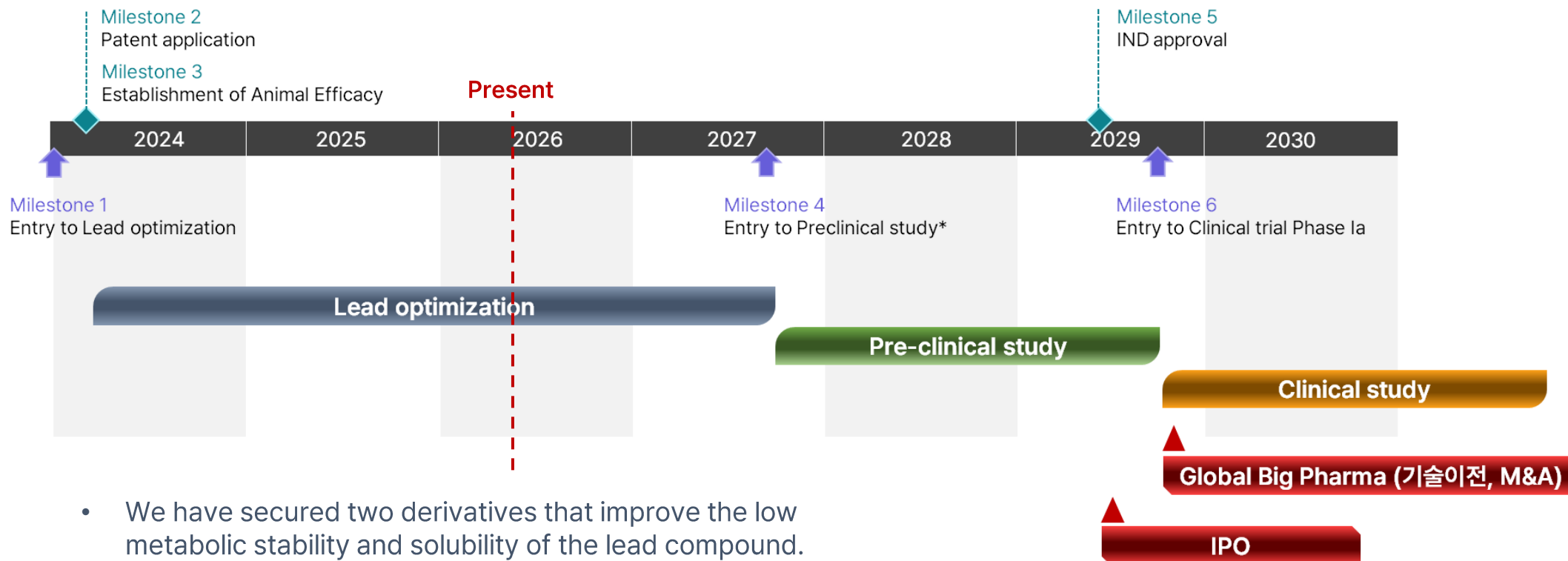


Restoration of hippocampal LTP





Project Timeline (Alzheimer's disease)

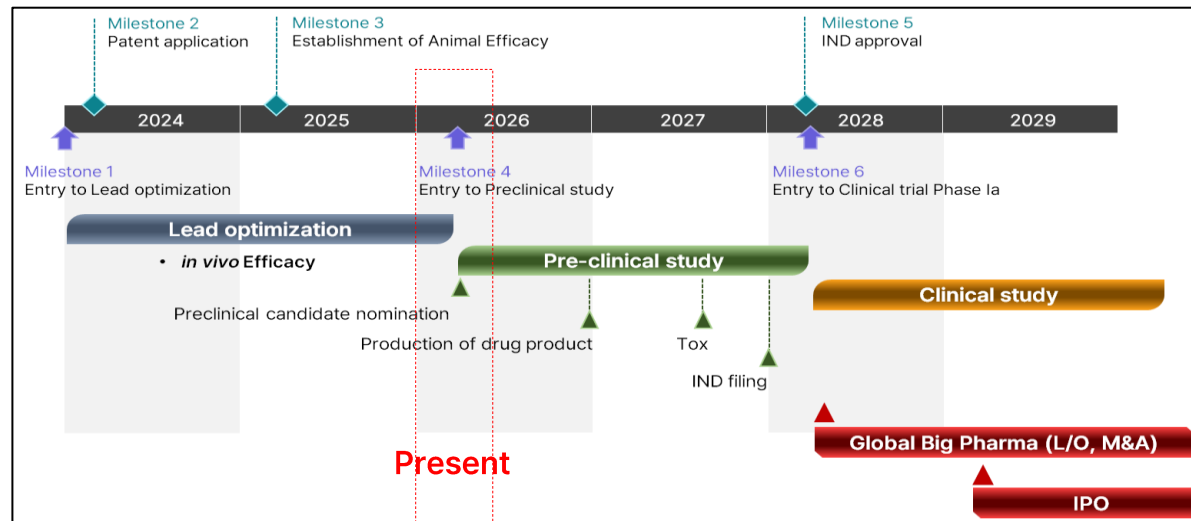


- We have secured two derivatives that improve the low metabolic stability and solubility of the lead compound.
- We have confirmed the in vivo efficacy of two derivatives in 5xFAD mice.



Investment Acquisition Plan

Project Timeline (Sarcopenia)



Project Timeline (Alzheimer's disease)



Series-A: \$3.5 millions

Series-B: \$7 millions

Marketing Goal and Strategy

Over 2 billion USD Licensing Deal via Targeted Strategy for Sarcopenia & Alzheimer's Markets

Goal

- 1st Goal: **2027** **Sarcopenia drug candidate** Signed CDAs & Initiated Due Diligence with 5+ Global Top 20 Pharmas
- 2nd Goal: **2028** **Sarcopenia drug candidate** **Licensing Deal over \$750 million USD Total Value (Upfront over \$75 million USD)**
- 3rd Goal: **2030** **AD drug candidate** **Licensing Deal over \$1.5 billion USD Total Value (Upfront over \$150 million USD)**

Strategy

- **Segmenting Big Pharma into 3 groups based on their needs for a tailored approach**

Group	Target Company	Needs & Engagement Strategy
Leaders in the obesity/metabolic disease & a strong commitment to AD drug development	- Eli Lilly - Novo Nordisk	[Obesity Combination Therapy] Value of Combination Therapy to Address Muscle Loss [AD] Suppressing amyloid production and restoring damaged neural networks [Sarcopenia+AD Dual Target] A platform technology enabling a single drug to target two indications
Leaders in aging and regenerative medicine	- Regeneron - Novartis - Sanofi	[Sarcopenia] Superior efficacy in improving muscle function compared to competing drugs [Sarcopenia+AD Dual Target] A platform technology enabling a single drug to target two indications
Leaders in CNS medicine	- Biogen - Eisai - Roche	[AD] Suppressing amyloid production and restoring damaged neural networks

● Our leadership team



CEO
Hyockman Kwon
Ph.D.

- Hankuk University of Foreign Studies (Korea), Professor
- Seoul National University (ROK), BA/MS
- Brandeis University (USA), Ph.D.
- University of Massachusetts Medical School (USA), Post-Doc
- Founder of KSMCB Epigenomics Academic Division



CTO
Yunhee Kim
Ph.D.

- Kyung Hee University (Korea), Professor
- Ewha Womans University (Korea), BA
- Seoul National University (Korea), MS
- Tufts University (USA), Ph.D.
- Harvard Medical School, Dana-Farber Cancer Institute (USA), Post-Doc
- Founder of KSMCB Neurodevelopmental Biology Academic Division



CSO
Sangmee Ahn
Ph.D.

- Seoul National University College of Pharmacy, B.S. M.S.
- MIT, Post-Doc
- Harvard Medical School Children's Hospital, Post-Doc
- Cambridge Neuroscience Inc., Staff Scientist
- Korea National Institute of Health, Director
- Dankook University College of Pharmacy, Dean



CFO
Andy Kim

Certified Investment Manager, AFPK

- Yonsei U. (BS), Korea U. (MS)
- Former VP, CFO of Vendit Inc.

● Scientific Advisors



Jung Min Joo
Ph.D.

Medicinal Chemistry

- Professor, Dep. Chemistry
Kyung Hee University



Young Ho Park
MD & Ph.D.

Clinical studies in Alzheimer's disease

- Professor, Dep. Neurology, Seoul National University Bundang Hospital

● Company History

- 2022. 01 Established ALIAD BIOPHARMA
- 2022. 01 Awarded 'Korea Dementia Research Center R&D Fund'
- 2022. 09 Awarded 'TIPS R&D Fund'
- 2023. 12 30 Health R&D Excellence Awards for 2023
- 2024. 08 Selected 'Adventure Challenge 1' by Huons Global
- 2024. 11 Signed 'Joint Research Agreement' with Shinpoong Pharm
- 2025. 04 Awarded 'National Drug Development Project' by KDDF
- 2026. 04 Awarded 'Unicorn Project, 1st stage'

● Partners



● Patents

- NOVEL COMPOUNDS WITH MTORC2-INHIBITORY ACTIVITY AND USES THEREOF
KR 10-2024-0039345 | PCT / KR2024 / 095625
- COMPOUNDS INHIBITING MTOR SIGNALING AND COMPOSITION FOR PREVENTING OR TREATING NEUROLOGICAL DISEASES COMPRISING THE SAME AS AN ACTIVE INGREDIENT
KR 102820816 | PCT / KR2022 / 002261
- SELECTIVE MTORC2 INHIBITOR AND USES THEREOF
KR 102816541 | PCT / KR2022 / 005189
- COMPOSITION FOR PROMOTING PROLIFERATION AND DIFFERENTIATION OF NEURAL STEM CELLS OR NEURAL PRECURSOR CELLS
KR 101097231B1



Redefining Healthy Aging

Aliad Biopharma Inc. is opening a new chapter in conquering aging.

Innovative drugs developed by a single-platform technology will restores muscle and brain functions and become the most reliable solution for ensuring a healthy aging.

Thank you.

CONTACT

Aliad Biopharma, Inc.
CEO, Hyockman Kwon, Ph.D.

✉ E-mail : hmkwon@aliadbiopharma.com

☎ Tel : +82-2-963-4553

📱 Mobile : +82-10-3288-4560

